Subsidies for truthful cooperation

Sharp proportional bounds and nonlinear escape

Maxwell Flanagan — Specialist review brief — 6 October 2026

Purpose. A focused request for correctness, priority and significance feedback on one manuscript. The results have written proofs, finite exact checks and internal model review. Historical novelty and human specialist validation remain unresolved.

The model. Fixed owners are partitioned into $k \geq 2$ complementary pools. Owner i reports a nonnegative capacity c_i; write

$$v_c(S) = \min_{1 \leq j \leq k} \sum_{i \in S \cap N_j} c_i, \qquad T = v_c(N).$$

A deterministic nonnegative payment rule must satisfy $y(S) \geq v_c(S)$ for every coalition and must be nondecreasing in each owner's own report. This is incentive compatibility for binding underreports when utility is payment and there are no use costs. A budget factor B means $y(N) \leq BT$; the subsidy is $(B-1)T$. Owners cannot change identities or pool memberships. Group, overreport and population-change guarantees are not assumed.

An optional floor is $y_i \geq H_i/k$, where H_i is the largest payoff in the *original efficient core*. The floor costs at most one extra budget factor via $\max\{y_i, [T - v_c(N \setminus \{i\})]/k\}$.

Main results and scope.

1. **All profiles: a sublogarithmic upper bound.** Set $d_i = \min(c_i, T)$ and let p_j be the probability that pool j has the smallest sum $\sum_{i \in N_j} d_i U_i$, for independent uniform U_i. The deterministic payment $d_i p_j$ covers all coalitions, is own-monotone, and has factor at most $R_k + 1/\log R_k$, where $R_k \log(R_k/e) = \log k$. This is $(1+o(1)) \log k / \log\log k$, uniformly over finite owner counts. It equals the established absolute continuous Banzhaf value of the normalized capped game, multiplied by T.

2. **Unrestricted necessary constant: two.** Any total-payment constant uniform over both pool and owner counts must satisfy $B \geq 2$. This is a necessary bound, not a factor-two construction or a divergence result.

3. **Proportional class: matching first-order growth.** For final payments $d_i \pi_j(c)$, with arbitrary nonnegative whole-profile pool prices, the floor-free minimax factor is $(1 + o(1)) \log k / \log\log k$. For r owners per pool an explicit certificate gives $B \geq kr/[k + (r-1)r^r]$.

4. **Nonlinear escape on the witnesses.** On all binary profiles in $\{1/r, 1\}$ with at most two altered pools, nonlinear payments have factor 2 (3 with the floor). On $[1/r, 1]$ with at most m altered pools, a different continuous rule has factor $2m + 3$ ($2m + 4$ with the floor). These are restricted-domain statements; the continuous formula's cost grows on an explicit family with arbitrarily many altered pools.

What is established in the literature. The underlying game and its nonempty core are established in Moretti–Norde, *A note on weighted multi-glove games* (2021), Definition 3.1 and Theorem 3.2; the final remark already covers real weights and discusses zeros. The integral functional is Kurz's absolute continuous Banzhaf index, arXiv:1312.5936v1, Definition 6.11. Neither ingredient is claimed as new.

The nearest strategic comparison is Huang–Mei–Yang–Cao, *Mechanism Design in Max-Flows* (2025), Sections 2.1–2.2 and 4.1–4.2. Serial parallel-edge layers realize our model as a special network. Their efficient truthful mechanisms need not cover every coalition. Our candidate results concern the cost of retaining that coverage across reports after relaxing efficiency.

Fixed-game cost of stability and population-cross-monotonic sharing are additional precedents, with different quantifiers. Our fixed games already have nonempty cores; any positive subsidy requirement here comes from linking allocations across reports. The manuscript and source ledger spell out those comparisons. This bounded source check is not a novelty certification.

The load-bearing distinction. The proportional lower proof transports a common pool price through report changes made by different owners. A nonlinear rule need not preserve that common price, and our witness family admits bounded nonlinear payments. Accordingly, the lower certificate does not prove that every truthful covering mechanism needs growing subsidy. The unresolved global question is

$$\text{Does one absolute } B < \infty \text{ work for all } k, \text{ owner counts, and nonnegative profiles?}$$

No polynomial-time exact evaluation theorem is claimed for the Banzhaf rule. The constants for the continuous restricted-domain rule are not claimed sharp.

Questions for a specialist.

1. Is there an existing theorem combining report monotonicity, coalition coverage and these budget guarantees, perhaps in continuous values, production games, resource monotonicity or truthful max-flow mechanisms?

2. Does known prior work already imply the first-order proportional-class bound, or the unrestricted necessary factor two?

3. Is the sharp class bound plus the explicit nonlinear escape a worthwhile standalone contribution? Would a different central theorem or framing make the paper more useful to the field?

4. Is there a specific gap in the capping/own-report argument, the linked inequality certificate, or the two-pool extension that should be repaired before wider circulation?

Suggested first reading. Manuscript Sections 2–3 (model and upper bound), Section 5 (sharp class bound), and Section 6 (escape). Section 4 supplies the unrestricted lower family; Section 7 and Appendix A give extensions. A brief assessment or a pointer to a missed theorem would be useful; a complete referee report is not requested.

Reproducibility. Six standard-library Python verifiers use exact arithmetic and explicit checks active under `-O`. Run `python3 -B -O verification/run_all.py` from the unpacked package. Finite checks supplement the proofs and do not establish priority. AI assistance in research, drafting and internal review is disclosed in the manuscript. External review remains pending; no endorsement is implied.